BY ORDER OF THE
SECRETARY OF THE AIR FORCE

AIR FORCE INSTRUCTION 13-101

18 FEBRUARY 2025

Command and Control Operations

EVALUATION OF GROUND BASED RADAR SYSTEMS AND SENSORS SUPPORTING COMMAND AND CONTROL (C2) ENTERPRISE

COMPLIANCE WITH THIS PUBLICATION IS MANDATORY

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This instruction implements Department of the Air Force Policy Directive (DAFPD) 13-1, *Command and Control (C2) Enterprise*. This instruction provides guidance and procedures for the Air Force program for evaluating ground-based surveillance radar systems and sensors (fixed, tethered and mobile). This publication applies to all civilian employees and uniformed members of the Regular Air Force, the Air Force Reserve, the Air National Guard, and those who are contractually obligated to comply with Department of the Air Force publications. This publication does not apply to the United States Space Force. This instruction requires the collection and or maintenance of information protected by the Privacy Act of 1974 authorized by Department of Defense Instruction (DoDI) 5400.11, *DoD Privacy and Civil Liberties Programs*. The applicable System of Records Notices (SORNs) F011 AF XO A, *Aviation Resource Management System* (ARMS) is available at: **https://dpcld.defense.gov/Privacy/SORNs/**. Ensure all records generated as a result of processes prescribed in this publication adhere to Air Force Instruction (AFI) 33-322, *Records Management and Information Governance Program*, and are disposed in accordance with (IAW) the Air Force Records Disposition Schedule (RDS), located in the Air Force Records Information Management System (AFRIMS). Refer recommended changes and questions about this publication to the office of primary responsibility (OPR) using the Department of the Air Force (DAF) Form 847, *Recommendation for Change of Publication*; route DAF Forms 847 from the field through the appropriate functional chain of command, to Air Combat Command (ACC)/Command and Control, Intelligence, Surveillance, Reconnaissance (C2ISR) Operations Division (ACC/A3C), 205 Dodd Blvd, Suite 121, Joint Base Langley-Eustis VA 23665-2789, ACC/A3C C2ISR Operations Division <**acca3c.c2isrops@us.af.mil**>. This publication may be supplemented at any level, but all direct supplements must be routed to the OPR of this

publication for coordination prior to approval. Major commands (MAJCOMs) are to forward proposed MAJCOM-level "direct supplements" to this volume to Air Force Training and Exercises Division (AF/A3TE), through ACC Combat Integration Branch (ACC/A3CI), **ACCA3YICombat@us.af.mil** for review and coordination prior to approval. The authorities to waive wing or unit level requirements in this publication are identified with a tier ("T-0, T-1, T-2, T-3") number following the compliance statement. See Department of the Air Force Manual (DAFMAN) 90-161, *Publications Processes and Procedures*, for a description of the authorities associated with the tier numbers. Submit requests for waivers through the chain of command to the appropriate tier waiver approval authority, as identified in **paragraph 1.3**, or alternately, to the publication OPR for non-tiered compliance items.

SUMMARY OF CHANGES

This document has been substantially revised and needs to be completely reviewed. The document was reorganized for enhanced clarity and to avoid duplication of information. Additionally, roles and responsibilities for support, funding, tasking, testing, and data release authority are now identified.

Chapter 1

OVERVIEW

1.1. General. Radars, sensors, and C2 systems play a vital role in air domain awareness. Many special types of radars are installed in numerous operational environments. The capability of each radar, sensor and system must be measured and adjusted so they can be configured to provide optimum performance and effectively meet their operational and technical mission requirements.

1.2. Scope. Air Force policy is to ensure tactical and strategic air defense ground-based radars, sensors, C2 systems, and electronic resources are installed, maintained, and operated in a high state of readiness to provide the ability to rapidly detect threats. Ground-based radar systems evaluation provides management with a tool for determining the capability of each radar system to detect, identify, track and control aircraft; determine aircraft height and maximize operational readiness and flight safety. In addition, ground-based radar systems evaluations determine the configuration for optimal performance in a dynamic weather and radar environment. Lastly, through continuous evaluation of radar systems, C2 systems are provided accurate information to process, track, and display data. These systems include Joint Surveillance System (JSS), North Warning System (NWS), Tethered Aerostat Radar System (TARS), Wide Area Surveillance (WAS) sensors, Atmospheric Early Warning System (AEWS), National Capital Region-Integrated Air Defense System (NCR-IADS), Air Defense Sector (ADS) Battle Control Center (BCC), the Control and Reporting Center (CRC), cloud-based command and control (CBC2) systems and Tactical Operations Centers (TOCs). The Air Force will evaluate these, and any other rapidly integrated systems and databases as needed in support of Great Power Competition preparedness.

1.3. Waivers.

1.3.1. ACC Director of Operations (ACC/A3), Pacific Air Forces (PACAF) Air and Cyberspace Operations (PACAF/A3/6), and United States Air Forces in Europe-Air Forces Africa Director of Operations (USAFE-AFAFRICA/A3/10) will serve as the respective MAJCOM OPR with waiver approval authority for all requests for waivers from tiered compliance statements in this instruction. Guidance on filing a copy of approved written waivers with this volume is provided in DAFMAN 90-161.

1.3.2. All waiver requests will be submitted in writing through channels to the appropriate MAJCOM OPR. **(T-2)**

Chapter 2

ROLES AND RESPONSIBILITES

2.1. ACC C2ISR Division (ACC/A3C). ACC/A3C is the executive agency for the United States Air Force (USAF) Radar Evaluation Program. ACC/A3C will:

2.1.1. Maintain the Department of Defense JSS and Air Force's radar systems evaluation capability.

2.1.2. Support other service, DoD agency, government, or non-government organization requests on a case-by-case basis. Approval authority is ACC/A3CI. ACC/A3CI will coordinate all requests with DoD Long Range Radar Joint Program Office (ACC/A3AJ) and ACC C2 Mission Systems Branch (ACC/A6OC).

2.1.3. ACC will establish a periodic evaluation program for mobile ground-based radar systems (e.g., AN/TPS-75). This program will be tailored to mission requirements, detected losses in system performance, and/or time since the last evaluation.

2.1.4. ACC/A3C delegates ACC/A3CI authority for subject matter experts (SME) in the management for this instruction.

2.2. ACC Combat Integration Branch. ACC/A3CI is responsible for C2 systems integrations. ACC/A3CI will:

2.2.1. Coordinate any changes to this publication with all office of collateral responsibility (OCRs) listed below:

2.2.1.1. ACC/A3AJ Long Range Radar Joint Program Office (JPO).

2.2.1.2. ACC/A6OC Command and Control (C2) Mission System Branch (ACC/A6OC).

2.2.1.3. Pacific Air Forces (PACAF)/A3/6

2.2.1.4. United States Air Force—Air Forces Europe/Africa (USAFE-AFAFRICA)/A3CO.

2.2.1.5. National Guard Bureau (NGB)/Air National Guard Operations (A3/10)

2.2.1.6. Air Forces Central Command current of operations (AFCENT/A3C)

2.2.2. Review all requirements for 84 Radar Evaluation Squadron (84 RADES) support with ACC/A3AJ and ACC/A6OC. **Note:** Radar systems that feed data into the North American Aerospace Defense Command (NORAD)/United States Northern Command (USNORTHCOM) enterprise will receive top priority over all other requirements.

2.2.3. Be the tasking authority for 84 RADES evaluation activity and will coordinate the evaluation schedule with ACC/A3AJ and ACC/A6OC.

2.3. Air Combat Command (ACC). As the combat air forces lead command, ACC is responsible for funding evaluations and evaluation support requirements to support fleet-wide program acquisition and modernization efforts.

2.4. ACC Resources and Budget (ACC/FM). ACC/FM will ensure funding is provided to support state-of-the-art radar evaluation technology, evaluations of proposed systems in the design review phase of acquisition, the maintenance/sustainment/improvement of radar evaluation networks, and other support requirements set by the SMEs.

2.5. Program Element Monitors (PEMs). PEMs for ground-based radar systems will ensure fleet-wide modification programs include funding to support necessary evaluations by the 84 RADES.

2.6. Air Force Personnel Center (AFPC). Due to the unique mission, specialized equipment, and specialized training required at the 84 RADES, AFPC will normally limit 84 RADES enlisted assignments to personnel who hold 7-level skill qualifications. Guidance to ensure an assignment availability code is annotated by the 84 RADES commander (84 RADES/CC) is provided in the ACC stabilized tour guides. ACC will ensure all positions defined by requirements set by the Air Force Manpower Analysis Agency (AFMAA) and their operational maturity model (OMM) are funded positions.

2.7. Major Commands (MAJCOMs). MAJCOMs operating ground-based surveillance radars, sensors and C2 systems will:

2.7.1. Program funds for the 84 RADES evaluation of systems under the MAJCOM commander's operational control and related flight support.

2.7.2. Provide flight support for evaluating command fixed, tethered and mobile radars, sensors and C2 systems.

2.7.3. Request services for baseline, special, and system radar evaluations according to **Chapter 5**.

2.7.4. When a fixed radar system is not being evaluated using 84 RADES automated remote recording and monitoring equipment, the using command must maintain a periodic radar evaluation program.

2.8. The 84 Radar Evaluation Squadron (RADES). 84 RADES is ACC's SME for the planning, development, testing and implementation of ground-based radar, sensors, C2 systems and related matters. 84 RADES will:

2.8.1. Conduct baseline, special, system evaluations and continuous performance monitoring and evaluation for ground-based surveillance radars, sensors and C2 systems.

2.8.2. Conduct radar site surveys, radar site analysis, and produce radar coverage products.

2.8.3. Conduct interference and obstruction analysis (e.g.,) man-made structures within system's line of sight) for ground-based radars and sensors for NORAD/NORTHCOM (N-NC) or as tasked by ACC/A3CI.

2.8.4. Extract relevant obstruction projects from the federal aviation administration (FAA) obstruction evaluation and airport airspace analysis system and conduct a technical analysis of the impact of obstructions within radar line of sight on the performance of long range and short-range radars/sensors used for N-NC missions.

2.8.5. Conduct analysis evaluations that include assessments of proposed and existing obstructions (such as a building or wind turbine farm) on the performance of the radar as well as determine the cumulative impact of the existing interference. The initial technical analysis

reports will be forwarded to N-NC and ACC/A3AJ for further action if the obstruction impacts the radar's performance.

2.8.6. Provide analysis products that will use a combination of radar, atmospheric, terrain/screening, and other supplementary data sets to provide a comprehensive product to DoD or other government agencies supporting track-of-interest, event analysis, aircraft accident investigations, search and rescue missions, and test and evaluation of ground-based radars and C2 sensors.

2.8.7. Support recorded radar/sensor data requests IAW **Chapter 5**.

2.8.8. Establish and maintain operating locations (OLs) at each of the N-NC ADS. OLs will confirm data integrity of radar and C2 sensors feeding the C2 systems or prior to introduction to N-NC enterprise; collects and analyzes radar and sensor performance data; and provides radar forensics in support of ADS missions. OLs will provide support to the ADS through foreign entity incursion and high priority event reconstructions, radar/sensor integration recommendations for priority airspace events, crew positional training on RADES equipment & software.

2.8.9. Maintain the capability to deploy survey and evaluation teams worldwide in support of contingency operations.

2.8.10. Develop and acquire specialized tools necessary to complete C2 system, radar and sensor evaluation missions.

2.8.11. Develop and maintain mission-specific infrastructure, including networks and information systems, at the appropriate classification level to support the software and hardware tools used in the radar, sensor, and C2 system evaluation mission.

2.8.12. Develop and maintain RADES Software (RS4) along with future RADES software versions, distributed radar analysis and monitoring software (DRAMS), and other software required to accomplish the squadron's mission.

2.8.13. Develop, maintain, and obtain an authority to operate (ATO) for special purpose networks, datacenters, and systems required to develop and operate mission-specific software and hardware at an operating classification level appropriate for the systems being evaluated.

2.8.14. Support approved agencies/units with RADESNet/special purpose advanced radar test and analysis (SPARTA) access, equipment, and training. Some units will not have full RADESNet/SPARTA access, instead they will have stand-alone radar data interface system (RDIS) equipment for radar evaluation and monitoring within their own networks.

Chapter 3

RADAR EVALUATION

3.1. Baseline Evaluation. This evaluation is performed to:

3.1.1. Determine optimum configuration of adjustable and selectable features (software and hardware), capabilities and limitations, and airspace coverage. It is tailored to individual radar sites and missions and includes effects of weather and terrain on detection and tracking performance. Dedicated flights may be required to make this assessment.

3.1.2. Provide a performance database and determine operational parameters and airspace coverage for inclusion in the 84 RADES electronic monitoring evaluation system.

3.2. Special Evaluation. This evaluation is performed to:

3.2.1. Assist an operating organization in isolating the causes of substandard performance.

3.2.2. Determine the effects of proposed and existing modifications on the operational capability and recommended configuration.

3.2.3. Determine the best site location or positioning of a radar and the optimum setting of adjustable features for use under varying conditions.

3.2.4. Determine the effects of proposed and existing structures within the radar line-of-sight on the operational capability and recommended configuration.

3.3. System Evaluation. This evaluation determines how well multiple sensors integrate into a central facility or system to include assessment of the radar inputs processing portion of the central system, such as radar message handling, tracker, and display functionality.

3.4. Continuous Performance Monitoring and Evaluation. This type of evaluation is a continuous near real-time evaluation process performed on previously baselined radars. The data collected during the baseline evaluation is used for initial pass/fail criteria and establishes a baseline for performance for each individual radar system. This evaluation is performed to:

3.4.1. Continuously monitor/assess radar performance based on short-term and long-term trends.

3.4.2. Continuously monitor radar data quality.

3.4.3. Detect radar performance deviations from established standards.

3.4.4. Alert users of system effectiveness and deterioration in detection capability.

3.4.5. Allow for corrective action to be affected before complete system failure.

3.5. Remote Evaluations. This type of evaluation uses recorded or live data feeds to perform a radar or C2 sensor system evaluation. It supplements the on-site evaluation through the use of available radar data feeds and specialized software to provide an in-depth look at a particular system's performance. This type of evaluation may fulfill the requirements of a re-baseline, special or system evaluation when conditions and/or equipment cannot support an on-site evaluation.

3.6. Baseline and Re-baseline. An initial baseline evaluation is required as soon as possible after a unit or agency completes installation acceptance testing and resolves major exceptions. The

radar should be operationally stable and properly connected to the C2 enterprise. A baseline evaluation and integration to the C2 enterprise is required before the radar is fully mission capable. Mobile radar systems may require a baseline evaluation if their deployed location becomes fixed.

3.6.1. Re-baseline and integration into the C2 system is necessary following major repairs, major modifications, or permanent radar relocation or as specified in Air Force Manual (AFMAN) 11-225 IP, *United States Standard Flight Inspection Manual.*

3.6.2. A baseline evaluation, conducted in conjunction with the FAA is required before the commission of radars in the JSS and/or the integration of air defense ground-based radars into the National Airspace System.

3.6.3. The 84 RADES will monitor performance of previously baselined fixed radars on a continuous basis, and they will perform a special evaluation when degradation is indicated either remotely or in-person at the radar site. **(T-2)**

3.6.4. Each established (in-place) radar or sensor system should receive an evaluation within 8 years, if possible, to assure sustained and optimal performance, as well as to validate the site's environmental impacts on the system.

Chapter 4

NEW SYSTEM FIELDING

4.1. Engineering Support. The 84 RADES will provide engineering level SMEs during the developmental testing (DT)/operational testing (OT) phases of radar acquisition and fielding and will evaluate new radar and C2 systems prior to integration into the C2 enterprise when verification services are requested by N-NC, other combatant commands, or allied countries and approved by ACC/A3CI. **(T-2)**

4.2. Test Agency Support. The 84 RADES supports test agencies/units by providing radar SMEs to support their tests. The 84 RADES is not a test organization. The following units are regularly supported by the 84 RADES:

4.2.1. 46th Test Squadron (46 TS) for DT.

4.2.2. 605th Test and Evaluation Squadron (605 TES) for OT.

4.2.3. Air Force Operational Test & Evaluation Center (AFOTEC) for OT.

4.2.4. FAA Technical Operations, Air Traffic Organization (FAA AJW) DT and OT events for new radars.

Chapter 5

SUPPORT REQUESTS

5.1. Requests. All requests for support from the 84 RADES will be submitted to ACC/A3CI. **(T-2)** Their email address is **ACCA3YICombat@us.af.mil**. ACC/A3CI will coordinate all requests with ACC/A3AJ and ACC/A6OC. **(T-2)** All requests are to be considered on a case-by-case basis with priority given to support N-NC missions. Other priorities are to be Air Force, other service or DoD agency, other government agency, or non-government agency, in that order. Prioritizations for requests are:

5.1.1. Priority 1: Operational systems (inoperable).

5.1.2. Priority 2: Operational system (degraded).

5.1.3. Priority 3: New program.

5.1.4. Priority 4: Operational system.

5.1.5. Priority 5: All others.

5.2. Classification Guidance. Classification will be IAW classification guides, such as Sensor/Radar Program Security Classification Guides and Sensor/Radars Security Classification References. The release authority for 84 RADES coverage products, evaluation reports, data analysis studies and other 84 RADES generated artifacts will be the 84 RADES/CC.

5.3. Recorded Radar/Sensor Data Requests. The 84 RADES supports requests for recorded radar/sensor data for use with test and evaluation, safety investigations or other specified situations to DoD or other government agencies as required. Request for release of any recorded radar data must be approved through ACC/A3AJ. **(T-2)**

5.3.1. ACC/A3AJ will maintain a historical archive of recorded sensors. The historical archive will include sensor data for not less than the previous seven years. **(T-2)**

5.3.2. Analysis products will include track-of-interest, event analysis, aircraft accident investigations, search and rescue missions for operating locations, safety centers and other government agencies. **(T-2)**

5.4. Radar Evaluation/Site Survey Requests. All radar evaluation/site survey requests will, as a minimum, contain the following information:

5.4.1. Purpose. Define problems and state the reason for the request. **(T-2)**

5.4.2. Location. **(T-2)**

5.4.3. Type of radar or C2 system and associated equipment to be evaluated. (**T-2)**

5.4.4. Desired date for support. **(T-2)**

5.4.5. Mission impact if the evaluation is not performed by the desired date. **(T-2)**

5.4.6. Special items of concern and other pertinent information. **(T-2)**

5.4.7. Points of contact. **(T-2)**

5.4.8. Funding source to conduct requested support. After being tasked, 84 RADES contacts the requesting unit to provide a cost estimate and to secure a fund cite. If applicable, include funds for dedicated flights if requestor is not providing aircraft. **(T-2)**

5.5. Engineering Support/Test Requests. All engineering support requests, including C2 testing, recorded radar data, live data feeds, network connectivity issues, and software support will, as a minimum, contain the following information:

5.5.1. Purpose. Define problems and state the reason for the request. **(T-2)**

5.5.2. Requesting agency and organization. **(T-2)**

5.5.3. Desired date for support. **(T-2)**

5.5.4. Mission impact if the evaluation is not performed by the desired date. **(T-2)**

5.5.5. Special items of concern and other pertinent information. **(T-2)**

5.5.6. Points of contact. **(T-2)**

5.5.7. Testing requests (OT, DT, research and development, analysis, etc.).

5.5.8. C2 system to be tested. **(T-2)**

5.5.9. Dates and location of testing. **(T-2)**

5.5.10. Listing of testing requirements. **(T-2)**

5.6. Recorded Data Requests. (e.g., National Transportation Safety Board Investigations, Air Force Safety Center investigations, legal request, obstruction analysis). All recorded data requests will, as a minimum, contain the following:

5.6.1. Start and stop date/time of requested data.

5.6.2. Release authorization by data owner(s). Request for release of any recorded radar data must be approved through ACC/A3AJ. If authorization has not been granted, coordinate with the 84 RADES to assist in identifying the proper release authority/data owner.

5.6.3. Area of interest (including top left and lower right coordinates, or center point and radius of the area of interest). Alternatively, the customer can provide a list of all radar sites to be included.

5.7. Software support requests: All recorded data requests will, as a minimum, contain the following:

5.7.1. Software title requested.

5.7.2. Description of support required.

5.7.3. Email address for software delivery.

5.8. Track of Interest, Event Analysis and Radar Data Reduction Request. All tracks of interest, event analysis and radar data reduction requests will contain the following information as known:

5.8.1. Description of the needed reduction including special items of concern and other pertinent information. **(T-2)**

5.8.2. Incident class (A-F or none). **(T-2)** (If unknown, coordinate with 84 RADES)

5.8.3. Priority (High/Med/Low). **(T-2)** (If unknown, coordinate with 84 RADES)

5.8.4. Requesting agency and organization. **(T-2)**

5.8.5. Desired completion date. **(T-2)**

5.8.6. Mission impact if the evaluation is not performed by the desired date. **(T-2)**

5.8.7. Points of contact name, organization/office symbol, phone number, and email. **(T-2)**

5.8.8. Product routing information to include name, organization/office symbol, phone number, and email. **(T-2)**

5.8.9. Incident details for each aircraft will include:

5.8.9.1. Aircraft type. **(T-2)**

5.8.9.2. Incident date. **(T-2)**

5.8.9.3. Start time (Z) and stop time (Z). **(T-2)**

5.8.9.4. Modes and codes and other broadcast identification if available. **(T-2)**

5.8.9.5. Latitude and longitude. **(T-2)**

5.8.9.6. Heading. **(T-2)**

5.8.9.7. Tail number. **(T-2)**

5.8.9.8. Departure airport. **(T-2)**

5.8.9.9. Arrival time. **(T-2)**

ADRIAN L. SPAIN, Lt Gen, USAF
Deputy Chief of Staff, Operations

Attachment 1

GLOSSARY OF REFERENCES AND SUPPORTING INFORMATION

References

Air Force Instructions (AFI) 33-322, *Records Management and Information Governance Program*, 23 March 2020

Air Force Manual (AFMAN) 11-225_IP, *United States Standard Flight Inspection Manual*, 1 April 2015

Air Force Policy Directive (DAFPD) 13-1, *Command and Control (C2) Enterprise*, 22 April 2021

Department of Air Force Manual (DAFMAN) 90-161, *Publications Processes and Procedures*, 18 October 2023

Department of Defense Directive (DODD) 5400.11, *DoD Privacy and Civil Liberties Programs*, 29 January 2019

System of Records Notices F011 AF XO A, *Aviation Resource Management System (ARMS)*, 4 March 2011

Prescribed Forms

None

Adopted Forms

Department of Air Force (DAF) Form 673 Publication/Form Action Request

Department of Air force (DAF) Form 847 Recommendation for Change of Publication

Abbreviations and Acronyms

84 RADES—84 Radar Evaluation Squadron

ACC—Air Combat Command

ADS—Air Defense Sector

AEWS—Atmospheric Early Warning System

AFCENT—Air Forces Central Command

AFI—Air Force Instruction

AFMAA—Air Force Manpower Analysis Agency

AFMAN—Air Force Manual

AFOTEC—Air Force Operational Test & Evaluation Center

AFPC—Air Force Personnel Center

AFRIMS—Air Force Records Information Management System

AJW—Technical Operation Services (Federal Aviation Administration Technical Operation Services)

ARMS—Aviation Resource Management System

ATO—authorization to operate

BCC—battle control center

C2—command and control

C2ISR—command and control, intelligence, surveillance, reconnaissance

CBC2—cloud based command and control

CRC—control and reporting center

DAF—Department of Air Force

DAFMAN—Department of Air Force Manual

DAFPD—Department of the Air Force Policy Directive

DoD—Department of Defense

DoDI—Department of Defense Instruction

DRAMS—distributed radar analysis and monitoring software

DT—developmental testing

FAA—Federal Aviation Administration

IAW—in accordance with

JSS—joint surveillance system

MAJCOM—major command

NCR-IADS—National Capital Region-Integrated Air Defense System

NGB—National Guard Bureau

N-NC—NORAD/NORTHCOM

NORAD—North American Aerospace Defense Command

NORTHCOM—Northern Command

NWS—north warning system

OCR—office of collateral responsibility

OL—operating location

OMM—operational maturity model

OPR—office of primary responsibility

OT—operational testing

PACAF—Pacific Air Forces

PEM—Program Element Monitor

RDIS—rada data interface system

RDS—records disposition schedule

SME—subject matter expert

SORN—system of records notices

SPARTA—special purpose advanced radar test and analysis

TARS—Tethered Aerostat Radar System

TES—test and evaluation squadron

TOC—tactical operations center

TS—test squadron

USAF—United States Air Force

USAFE-AFAFRICA—United States Air Forces in Europe/Air Forces Africa

WAS—wide area surveillance

Office Symbols

84 RADES/CC—84th Radar Evaluation Squadron Commander

ACC/A3—ACC Director of Operations

ACC/A3AJ—ACC Long Range Radar Joint Program Office

ACC/A3C—ACC Command, Control, Intelligence, Surveillance, and Reconnaissance Operations Division

ACC/A3CI—ACC Combat Integration Branch

ACC/A6OC—ACC C2 Mission Systems Branch

ACC/FM—ACC Resources and Budget

AF/A3T—Air Force All Domain Command and Control Operations

AF/A3TE—Air Force Exercises and Training Division

AFCENT/A3C—Air Force Central Command Current Operations

FAA AJW—Federal Aviation Administration Technical Operations Services, Air Traffic Organization

PACAF/A3/6—Pacific Air Forces Air and Cyberspace Operations

NGB/A3/10—Air National Guard Operations

NGB/A3C—Air National Guard C2ISR & Airfield Operations

USAFE-AFAFRICA/A3/10—United States Air Forces in Europe-Air Forces Africia Director of Operations, Strategic Deterrence and Nuclear Integration